Contribution Journal: Lab 05 – Data Preparation

Course: ITAI-1371 Machine Learning

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Arinze Okafor

1. Role Overview & Leadership Responsibilities

For Lab 05, I functioned as the Data Lead and Technical Coordinator for the group. In this role, my primary objective was to ensure the integrity of the data preprocessing pipeline from initial ingestion through feature scaling, while coordinating the project workflow and synthesizing individual contributions into a unified submission.

My responsibilities spanned three primary domains:

- Team Follow-Up & Coordination: Actively following up with team members to monitor task progress, clarifying requirements from the Lab 05 Data Preparation instructions, and establishing internal checkpoints to ensure alignment prior to final assembly.
- Data Synthesis & Assembly: Collecting intermediate outputs, resolving code inconsistencies across tasks, consolidating the preprocessing pipeline into a single Jupyter notebook, and structuring the deliverable repository.
- Core Technical Execution & Quality Assurance: Implementing critical data transformation tasks, debugging deprecation warnings, validating mathematical correctness, and answering theoretical knowledge checks.

2. Detailed Breakdown of Individual Contributions

Task Coordination & Team Management

- Initiated group communication to delineate responsibilities across missing-value analysis, categorical feature transformation, and numerical scaling.
- Conducted scheduled follow-ups with team members to ensure all assigned sections adhered to the lab rubrics and grading standards.

- Collected individual code segments and conceptual inputs, identifying discrepancies in data types and naming conventions before merging them into the master notebook.

3. Verification & Submission Management

- Executed end-to-end execution testing using Google Colab's Restart session and run all to ensure deterministic execution with zero runtime errors.
- Prepared and verified the final PDF exports in accordance with the required course naming standards: o L05_YourGroupName_ITAI1371.pdf o L05Journal_R_YourGroupName_ITAI1371.pdf o L05Journal_C_YourGroupName_ITAI1371.pdf
- Organized the repository folder Lab-05-Data-Preparation/ in GitHub and confirmed the directory link for Canvas submission.

John’a Matthews

Lab 5 – Contribution Journal

John’a Matthews

Introduction

For Lab 5, my contribution focused on preparing the Titanic dataset for machine learning. I worked through the data preparation tasks, wrote Python code, reviewed the output, and considered why each preprocessing technique is useful. The lab helped me understand that data often needs to be cleaned and converted before a machine learning model can be used.

My Specific Contributions

First, I handled missing values in the Age column. I calculated the median age and used the fillna() method to replace missing entries. I then checked the column with isnull().sum() to verify that no missing Age values remained. I chose the median because it is less affected by extreme values than the mean.

Next, I used pd.get_dummies() to encode the Sex and Embarked columns. I set drop_first=True and saved the result in encoded_df. This converted the categorical information into numerical indicator columns and removed the original columns that were encoded.

I also applied feature scaling to the Age and Fare columns. I imported StandardScaler, created a scaler, selected the two numerical columns, and used fit_transform() to standardize their values. Finally, I displayed the first few rows to inspect the data that was transformed. I completed the knowledge check by explaining median imputation, one-hot encoding, and why decision trees usually do not require feature scaling.

My Learning and Reflection

This lab helped me connect data preparation methods to the machine learning workflow. I learned that imputing missing values can preserve rows that might otherwise be deleted, while one-hot encoding makes categorical information usable by models that require numerical input. I also learned that standardization puts numerical features on a comparable scale, which can be useful for some algorithms. The lab showed me that preprocessing choices depend on the dataset and the model being used, rather than every technique being necessary in every situation.

Limitations and Future Contribution

In future projects, I would improve my contribution by checking all columns for missing values and deciding how to handle each one, rather than focusing only on Age. I would also verify the data types of the encoded columns and compare model results before and after scaling when appropriate. For a real predictive task, I would fit preprocessing steps using the training data only to avoid data leakage.

Conclusion

Overall, my contribution to Lab 5 was to complete and review the data preparation steps for the Titanic dataset. I practiced filling missing values, encoding categorical features, scaling numerical columns, and explaining the purpose of these techniques. This work strengthened my Python data-analysis skills and helped me understand how thoughtful preprocessing supports machine learning.

Hamza Ajmal

Module 05 Lab – Contribution Journal

ITAI-1371: Introduction to Machine Learning | Data Preparation

Hamza Ajmal

Individual contribution

This journal records the work I completed while working through the Module 05 Data Preparation lab. I inspected the notebook, completed the student code sections, and reviewed the outputs produced by the Titanic dataset workflow. The statements below should be edited if any part does not accurately describe my personal work.

Initial inspection and setup

I ran the setup section to load the Titanic dataset and inspect the first rows and missing-value counts. The output showed 177 missing Age values, 687 missing Cabin values, and 2 missing

Embarked values. I also addressed a syntax issue in the setup cell that had joined separate Python statements together, which prevented the dataframe from being created.

Missing-value imputation

I completed the Age imputation task by calculating the median Age, filling missing Age values with that median, and checking the result. The verification output reported zero missing values in Age. I kept the task focused on the column requested by the lab rather than claiming that every missing value in the dataset had been removed.

Encoding and scaling

I completed the one-hot encoding task for Sex and Embarked and checked the first rows of the resulting dataframe. The output included Sex_male, Embarked_Q, and Embarked_S. I then used StandardScaler on Age and Fare and reviewed the transformed sample to confirm that the values were standardized.

Verification and learning

I checked the printed outputs after each task and learned that notebook cells depend on variables created by earlier cells. When the setup cell failed, later cells could not find df or encoded_df. Correcting the setup code and running the notebook in order resolved those downstream errors. I also reviewed why median imputation, one-hot encoding, and scaling are used, and why preprocessing should be fitted on training data only in a real modelbuilding workflow.

Jonathon Wilturner

Lab 5 – Contribution Journal

Jonathon Wilturner

Introduction

For Lab 5, my contribution focused on preparing the Titanic dataset for machine learning. I worked through the data preparation tasks, wrote Python code, reviewed the output, and evaluated why each preprocessing technique matters. The lab reinforced that data usually needs to be cleaned and converted before a machine learning model can use it.

My Specific Contributions

First, I handled missing values in the Age column. A check with isnull().sum() showed 177 missing entries out of 891 rows. I calculated the median age of 28.0 and used fillna() to replace the missing values. I then ran isnull().sum() again to confirm that zero missing Age values remained. I chose the median because Age is skewed and the median is less affected by extreme values than the mean.

Next, I used pd.get_dummies() to encode the Sex and Embarked columns with drop_first=True and saved the result in encoded_df. This replaced the original text columns with numerical indicator columns.

I also applied feature scaling to Age and Fare. I imported StandardScaler, created a scaler, selected the two numerical columns, and used fit_transform() to standardize them. I displayed the first few rows to confirm the transformation worked. Finally, I completed the knowledge check by explaining median imputation, one-hot encoding, and why decision trees usually do not require feature scaling.

My Learning and Reflection

This lab connected data preparation methods to the machine learning workflow. Imputation preserves rows that would otherwise be deleted, which here meant keeping nearly 20% of

the dataset. One-hot encoding makes categorical information usable by models that require numerical input. Standardization puts features with very different ranges, like Fare and Age, on a comparable scale for algorithms that are sensitive to magnitude. Preprocessing choices depend on the dataset and the model, and not every technique is needed every time.

Limitations and Future Contribution

In future projects, I would check every column for missing values and decide how to handle each one, rather than focusing only on Age. For example, Embarked has 2 missing values and Cabin has 687. I would also verify the data types of the encoded columns, since pandas 2.x returns True and False values by default. Where appropriate, I would compare model results before and after scaling. For a real predictive task, I would fit preprocessing steps on the training data only to prevent data leakage.

Conclusion

My contribution to Lab 5 was completing and reviewing the data preparation steps for the Titanic dataset. I practiced filling missing values, encoding categorical features, scaling numerical columns, and explaining the purpose of each technique. This work strengthened my Python data analysis skills and showed me how deliberate preprocessing supports reliable machine learning.